UCLouvain

LINMA2222 - Stochastic Optimal Control & RL

Project - Part 4: Mean Poisson Error

Optimal Portfolio Strategy

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Question 8.1

Compare the Poisson error with the Bellman error seen in the course for deterministic systems and for stochastic systems with discounted cost.

Answer

Given a policy π and approximations Q^θ and η^θ, the Poisson error is given by:

$$\mathcal{P}^\theta(x_t, u_t) := \mathbb{E}\big[r_t + Q^\theta(x_{t+1}, \pi(x_{t+1})) \mid x_t, u_t\big] - \eta^\theta - Q^\theta(x_t, u_t)$$

where r_t is the stage reward, η^θ is an approximation of the average reward per stage and $Q^\theta(\cdot,\cdot)$ is an approximation of the **relative** Q-function.

Alternatively, the Bellman error is given by:

$$\mathcal{B}^\theta(x_t, u_t) := \mathbb{E}\big[r_t + \gamma Q^\theta(x_{t+1}, \pi(x_{t+1})) \mid x_t, u_t\big] - Q^\theta(x_t, u_t)$$

with $\gamma \in [0, 1)$ the discount factor.

- **Deterministic Systems**
  Given a deterministic system of the following form:
  $$x_{t+1} = f(x_t, u_t)$$
  The Poisson error (PE) is then given by:
  $$\mathcal{P}^\theta(x_t, u_t) = r_t + Q^\theta(f(x_t, u_t), \pi(f(x_t, u_t))) - \eta^\theta - Q^\theta(x_t, u_t)$$
  and the Bellman error (BE) by:
  $$\mathcal{B}^\theta(x_t, u_t) = r_t + \gamma Q^\theta(f(x_t, u_t), \pi(f(x_t, u_t))) - Q^\theta(x_t, u_t)$$
  The main differences are that PE has the $-\eta^\theta$ term and uses $\gamma = 1$.
- **Stochastic Systems**
  For stochastic with discount γ, PE is **not** valid. In fact, the Poisson equation assumes an ergodic average-reward setting as well as $\gamma = 1$. In this case, BE must be used.

In the context of our project, we want to maximize the infinite-horizon average reward. Because PE measures how well our approximations Q^θ and η^θ are, this is the kind of problem where we'd want to use it instead of BE.

Question 8.2

Show that if Q^{θ_1} and η^{θ_2} are linearly parametrized (i.e., $Q^{\theta_1}(x, u) = \theta_1^\top \psi_Q(x, u)$ for some $\psi_Q : \mathcal{X} \times \mathcal{U} \to \mathbb{R}^d$ and $\eta^{\theta_2} = \theta_2$), then minimizing the MSPE can be formulated as a linear system of equations to be solved in the least-square sense. Compute the matrix A_{lspe} and the vector b_{lspe} of the corresponding normal equations. Discuss the differences with the LSTD framework used in Part III.

Answer

In this setting, given a policy π, PE is given by:

$$\mathcal{P}^\theta(x, u) := \mathbb{E}\big[r + Q^{\theta_1}(x^+, \pi(x^+)) \mid x, u\big] - \eta_2^\theta - Q^{\theta_1}(x, u)$$

where $Q^{\theta_1}(x, u) = \theta_1^\top \psi_Q(x, u)$ and $\eta_2^\theta = \theta_2$.

Answer (cont.)

If we define the *expected next feature vector* as:

$$\overline{\psi}(x,u) := \mathbb{E}\big[\psi_Q(x^+, \pi(x^+)) \mid x, u\big]$$

and the expected reward as:

$$\overline{r}(x,u) := \mathbb{E}[r \mid x, u]$$

Then:

$$\mathbb{E}\big[r + Q^{\theta_1}(x^+, \pi(x^+)) \mid x, u\big] = \overline{r}(x,u) + \theta_1^\top \overline{\psi}(x,u)$$

and PE becomes:

$$\mathcal{P}^\theta(x,u) = \overline{r}(x,u) + \theta_1^\top \overline{\psi}(x,u) - \theta_2 - \theta_1^\top \psi_Q(x,u)$$

As suggested in the hint, we will define the following augmented vector:

$$\theta := [\theta_1 \;\; \theta_2]^\top$$

and define also:

$$\Psi(x,u) := \begin{bmatrix} \overline{\psi}(x,u) - \psi_Q(x,u) \\ -1 \end{bmatrix}$$

$$\Longrightarrow \mathcal{P}^\theta(x,u) = \overline{r}(x,u) + \theta^\top \Psi(x,u)$$

We thus want to solve:

$$\begin{aligned}
&\min_\theta \mathbb{E}_{(x,u)\sim\mu}\Big[\big(\overline{r}(x,u) + \theta^\top \Psi(x,u)\big)^2\Big] := \mathrm{L}(\theta) \\
&\longrightarrow \nabla_\theta \mathrm{L}(\theta) = 2\mathbb{E}_{(x,u)\sim\mu}[\Psi(x,u))(\overline{r}(x,u) + \theta^\top \Psi(x,u))] = 0 \\
&\Longleftrightarrow \underbrace{\mathbb{E}_{(x,u)\sim\mu}[\Psi(x,u)\Psi^\top(x,u)]}_{A_{\text{lspe}}}\theta = \underbrace{-\mathbb{E}_{(x,u)\sim\mu}[\Psi(x,u)\overline{r}(x,u)]}_{b_{\text{lspe}}}
\end{aligned}$$

The main difference with the previous LSTD framework is that we obtain an explicit estimation of the average reward η via θ_2 obtained by solving the the regression problem above.

Question 8.3

Propose an approximation of A and b in Question 8.2 from data obtained from the system controlled by some exploration policy π_{exp}, assuming that μ is the steady-state measure of the system controlled by π_{exp}. Then, explain how such a subroutine can be implemented if you have the transition kernel $\kappa(x' \mid x, u)$ of the system.

Answer

To approximate each quantity, we will first need to make ourselves a dataset. We will create the dataset as follows:

$$\{(x_i, u_i, r_i, x_i')\}_{i=1}^N$$

where:

- $(x_i, u_i) \sim \mu$
- $x_i' \sim \kappa(\cdot \mid x_i, u_i)$
- r_i is the received reward

Answer (cont.)

To obtain the steady-state samples in practice, we run a long trajectory under π_{exp}, discards an initial period and take samples spaced apart to reduce correlation.
Then, we have to be able to evaluate:

$$\overline{\psi}(x, u) = \mathbb{E}\big[\psi_Q(x^+, \pi(x^+)) \mid x, u\big]$$

We can do that under the stated assumption that we have access to a subroutine to compute:

$$\mathbb{E}[f(x^+, \pi(x^+)) \mid x, u] = \int f(x^+, \pi(x^+))\kappa(x' \mid x, u)\,\mathrm{d}x'$$

where f would be $\overline{\psi}$ in our case. To implement this subroutine, we can either compute it analytically if we are in the special case where κ has a closed-form expression for its primitive, or if it is not possible, we can use a Monte-Carlo approximation:

1. Given (x, u), sample M next states $x'^{(1)}, ..., x'^{(M)} \sim \kappa(\cdot \mid x, u)$
2. Compute:

$$\overline{\psi}_{\text{approx}}(x, u) = \frac{1}{M}\sum_{m=1}^{M}\psi_Q\big(x'^{(m)}, \pi\big(x'^{(m)}\big)\big)$$

To compute an approximation of A and b, we need to do that for each sample (x_i, u_i):

$$\longrightarrow \quad \overline{\psi}^{(i)}_{\text{approx}} = \frac{1}{M}\sum_{m=1}^{M}\psi_Q\Big(x_i'^{(m)}, \pi\Big(x_i'^{(m)}\Big)\Big)$$

We then construct $\hat{\Psi}^{(i)}$:

$$\longrightarrow \quad \hat{\Psi}^{(i)} = \begin{pmatrix} \overline{\psi}^{(i)}_{\text{approx}} - \psi_Q(x_i, u_i) \\ -1 \end{pmatrix} \in \mathbb{R}^{d+1}$$

Finally, we can construct $\hat{A}$ and $\hat{b}$:

$$\hat{A} = \frac{1}{N}\sum_{i=1}^{N}\hat{\Psi}^{(i)}\big(\hat{\Psi}^{(i)}\big)^\top$$

$$\hat{b} = -\frac{1}{N}\sum_{i=1}^{N} r_i\hat{\Psi}^{(i)}$$

where we assumed that we have access to r_i to act as an estimate for $\overline{r}(x_i, u_i)$.

Question 8.4

Provide a finite-support kernel that approximates the transition kernel of the system

Answer

Given a current state and control (x, u), the next state $x^+ = \begin{pmatrix} q^+ & z^{a+} & z^{u+} \end{pmatrix}^\top$ is given by:

$$\begin{cases} q^+ &= q + u & \text{(deterministic)} \\ z^{a+} &= (1-\omega^a)z^a + \omega^a\sigma^a\xi^a, & \xi^a \sim \mathcal{N}(0,1) \\ z^{u+} &= (1-\omega^u)z^u + \omega^u\beta^a u & \text{(deterministic)} \end{cases}$$

Answer (cont.)

To approximate the transition kernel, we will use a Monte-Carlo approximation. For each (x, u), we can:

1. Sample M independent noise values $\xi^{a,(1)}, ..., \xi^{a,(M)} \sim \mathcal{N}(0, 1)$
2. Compute $z^{a+,(m)} = (1 - \omega^a) z^a + \omega^a \sigma^a \xi^{a,(m)}$
3. $p_m(x, u) = 1/M$

The kernel is then given by:

$$\kappa_M(x^+ \mid x, u) = \frac{1}{M} \sum_{m=1}^{M} \delta_{x^{(m)+}(x,u)}(x^+)$$

where:

$$x^{(m)+}(x, u) = \begin{pmatrix} q + u \\ (1 - \omega^a) z^a + \omega^a \sigma^a \xi^{a,(m)} \\ (1 - \omega^u) z^u + \omega^u \beta^a u \end{pmatrix}$$

Question 8.5

Apply the LSPE algorithm to estimate the Q-function of the closed-loop system with policy π_{cl}. Use Question 8.4 to approximate the transition kernel of the system. To generate your data, use an exploration policy π_{exp}, e.g., use $\pi_{\text{exp}(\cdot \mid x)} \sim \mathcal{N}\left(K_{\text{cl}} x, \sigma_{\text{exp}}^2\right)$. Make also a reasonable choice of basis functions, motivated briefly. Comment your results.

Answer

For the choice of the basis, we chose the following one:

$$\psi(x, u) = \begin{pmatrix} q & z^a & z^u & u & q^2 & (z^a)^2 & (z^u)^2 & u^2 & q z^a & q z^u & q u & z^a z^u & z^a u & z^u u \end{pmatrix}^\top$$

The reason why we chose this is because it allows to capture non-linearities while still being low-dimensional enough for fast computation. In the previous part, it also allowed to perfectly approximate the LQR Q-function and we will thus use it again in this part.

After we implemented the LSPE algorithm previously described on the closed-loop system with policy π_{cl}, we plotted Q_{LSPE} learned from the algorithm against an empirical evaluation of the Q-function:

Answer (cont.)

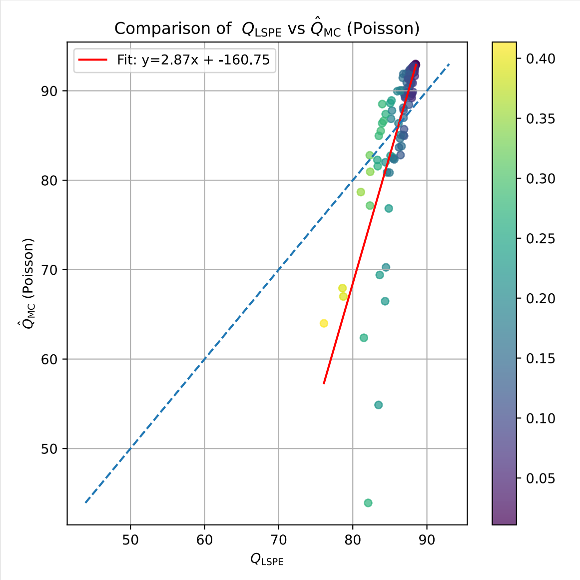

Figure 1: Experimental Q-function obtained via Monte-Carlo approximation against Q_{LSPE}

On Figure 1, we see that the Q_{LSPE} approximates the estimation quite well, though there is a slight deviation with the identity line.

Question 8.6

Consider the LQR approximate model obtained in Section 4, i.e., with objective (7). For this model repeat Question 8.5 (but for the comparison, use the exact Q-function, which you can compute analytically).

Answer

Again, we ran the LSPE algorithm, this time on the LQR approximate model from section 4. We use the same choice of basis as the last question.

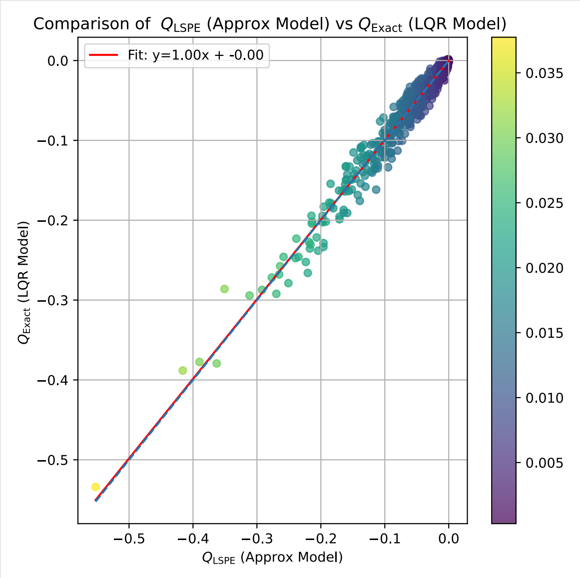

Figure 2: Exact Q-function of the LQR approximate model against Q_{LSPE}

On Figure 2, we can see that the obtained Q-function approximation approximates perfectly the exact Q-function of the approximate LQR model.

Question 8.7

For the setting of Question 8.5, implement the LSPE+PI algorithm, using π_{cl} as initial policy. Denote the resulting policy by π_{lspepi}. Repeat Question 3.2 for π_{lspepi}. Compare the averaged reward with that of the policies π_{cl} and π_{lqr}.

Answer

As stated in the assignment, we took the LSPE algorithm implemented in Question 8.5 and added a policy improvement step to get the LSPE+PI algorithm.

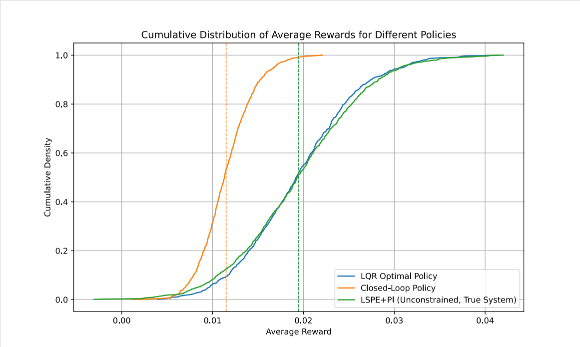

Figure 3: Cumulative empirical distribution of the average rewards for the different policies

On Figure 3, we can see that the π_{lspepi} policy performs exactly like the LQR one. This indicates that it performs quite well. We also see that it performs well better than the initial π_{cl}, which is expected for a policy *improvement* algorithm. Below is a plot of the empirical distribution of the average rewards to better see the performance of those policies:

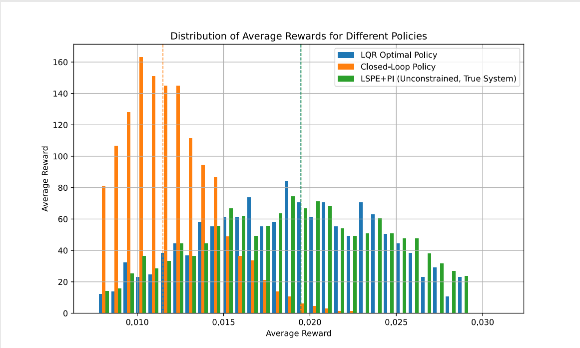

Figure 4: Empirical distribution of the average rewards for the different policies

Again, we see that the π_{lspepi} policy performs better than π_{cl}.

Question 8.8

For the setting of Question 8.6, implement the LSPE+PI algorithm, using π_{cl} as initial policy. Demonstrate the convergence of the gain sequence K_k to K_{lqr} (where K_{lqr} is the gain associated to π_{lqr}) by plotting the evolution of the error norm with k.

Answer

Again, we ran LSPE+PI but on the LQR approximate model starting with π_{cl} as an initial policy. As stated, we plotted the evolution of the error between K_k and K_{lqr}.

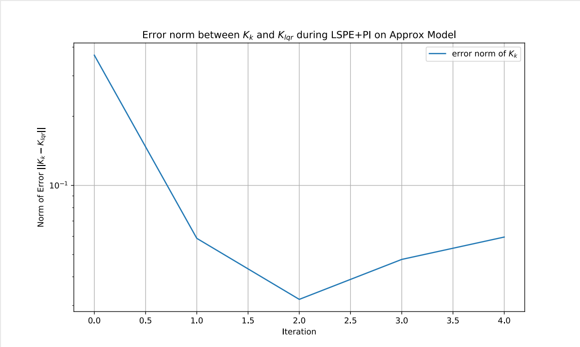

Figure 5: Convergence of K_k to K_{lqr}

Here on Figure 5, we see that after only 2 iterations, the LSPE policy converges to the optimal K_{lqr} one. Altough there is a small error even after convergences, the error is small enough for this approximated policy to be considered a good approximation.

As a comparison with the previous policies, we also plotted the distributions of the average reward:

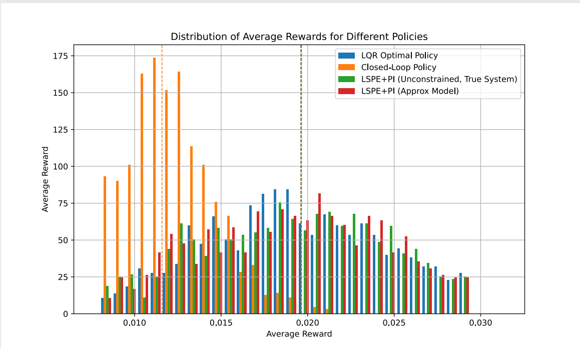

Figure 6: Empirical distribution of the average rewards for the different policies

We see that this policy performs exactly like the LQR optimal policy as well as the π_{lspepi} policy.

Question 8.9

Repeat Question 8.7 (i.e., using the true reward model of the system) but with the constrained policy improvement.

Answer

The policies may violate the constraint $q \in [0, 1]$, we thus solve the following constrained optimization problem during the policy improvement step:

$$\pi(x) = \arg\max_{u} Q^{\theta}(x, u) \quad \text{s.t.} \quad 0 \leq (1 \;\; 0 \;\; 0)x + u \leq 1$$

After implementing that, we compared the average reward distributions of all the previous policies:

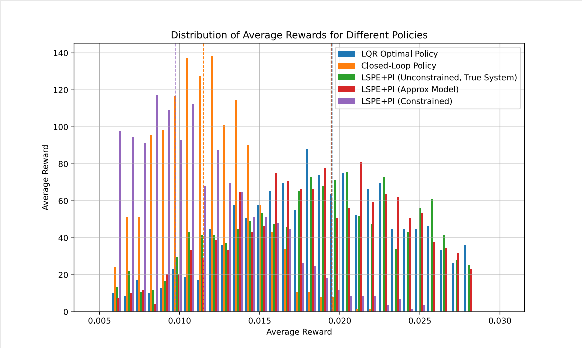

Figure 7: Empirical distribution of the average rewards for the different policies

On Figure 7, we see that the constrained policy is actually worse than the other ones. To get a fair comparison, we will plot them all again but now with the constraint applied:

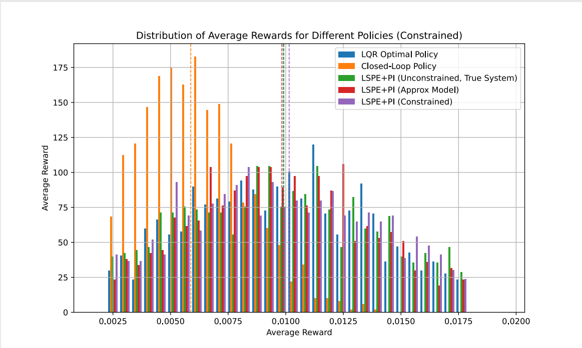

Figure 8: Empirical distribution of the average rewards for the different constrained policies

We now see that the policy we obtained in this question (purple distribution) is better than the other ones after we applied the constrained.